In [ ]:

```
# EXPERIMENT NO. 3: Statistical Analysis and Hypothesis Testing

# Step 2: Import Required Libraries        Pragya Singhal , Cu24220050 , 43
import pandas as pd
import numpy as np
import scipy.stats as stats
import statsmodels.api as sm
import statsmodels.formula.api as smf
import matplotlib.pyplot as plt
import seaborn as sns

# Set style for visualizations           Pragya Singhal , CU24220050 , 43
sns.set(style="whitegrid")
# Step 3: Load the Dataset and Calculate Descriptive Statistics
df = pd.read_csv("Housing.csv")

print("--- 1. DATASET HEAD ---")
print(df.head(), "\n")
print("--- 2. DESCRIPTIVE STATISTICS (Numerical Features) ---")
desc_stats = df.describe()
print(desc_stats, "\n")
for col in ['price', 'area']:
    print(f"Statistics for {col}:")
    print(f"  - Mean: {df[col].mean():.2f}")
    print(f"  - Median: {df[col].median():.2f}")
    print(f"  - Mode: {df[col].mode().values[0]}")
    print(f"  - Variance: {df[col].var():.2f}")
    print(f"  - Standard Deviation: {df[col].std():.2f}\n")

# Step 4:(Correlation Matrix & Heatmap)     Pragya Singhal , Cu24220050 , 43
print("--- 3. CORRELATION ANALYSIS ---")
numerical_df = df.select_dtypes(include=[np.number])
correlation_matrix = numerical_df.corr()
print(correlation_matrix[['price']])

plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm", fmt=".2f", linewidt
plt.title("Correlation Matrix Heatmap - Housing Dataset")
plt.show()


# Step 5 & 6: Formulate Hypothesis & Perform Independent Sample t-Test
print("--- 4. INDEPENDENT SAMPLE T-TEST ---")
basement_yes = df[df['basement'] == 'yes']['price']
basement_no = df[df['basement'] == 'no']['price']

t_stat, p_val = stats.ttest_ind(basement_yes, basement_no, equal_var=False)
print(f"T-statistic: {t_stat:.4f}")
print(f"P-value: {p_val:.4e}")

if p_val < 0.05:
    print("Conclusion: Reject H0. There is a statistically significant differenc
else:
    print("Conclusion: Fail to reject H0. No significant price difference found


# Step 7: Apply One-Way ANOVA     Pragya Singhal , Cu24220050 , 43
print("--- 5. ONE-WAY ANOVA ---")
```

```python
# Group prices by number of stories
groups = [group['price'].values for name, group in df.groupby('stories')]
f_stat, anova_pval = stats.f_oneway(*groups)

print(f"F-statistic: {f_stat:.4f}")
print(f"P-value: {anova_pval:.4e}")

if anova_pval < 0.05:
    print("Conclusion: Reject H0. There is a statistically significant differenc
else:
    print("Conclusion: Fail to reject H0. No significant difference across story

# Step 8 & 9: Build Simple Linear Regression Model      Pragya Singhal , Cu24220
print("--- 6. SIMPLE LINEAR REGRESSION MODEL ---")
model = smf.ols('price ~ area', data=df).fit()
print(model.summary())

# Visualizing the Regression Line     Pragya Singhal , Cu24220050 , 43
plt.figure(figsize=(8, 5))
sns.regplot(x='area', y='price', data=df, scatter_kws={'alpha':0.3}, line_kws={'
plt.title("Simple Linear Regression: House Price vs. Area")
plt.xlabel("Area (sq. ft)")
plt.ylabel("Price")
plt.show()

# Step 10: Summary of Findings     Pragya Singhal , Cu24220050 , 43
print("--- 7. SUMMARY OF FINDINGS ---")
print(f"- R-squared (R²): {model.rsquared:.4f} ({model.rsquared*100:.2f}% of pri
print(f"- Coefficient for Area: {model.params['area']:.2f} (For every 1 sq.ft in
print(f"- Model p-value (F-statistic p-value): {model.f_pvalue:.4e}")
```

```
--- 1. DATASET HEAD ---
      price  area  bedrooms  bathrooms  stories mainroad guestroom basement  \
0  13300000  7420         4          2        3      yes        no       no
1  12250000  8960         4          4        4      yes        no       no
2  12250000  9960         3          2        2      yes        no      yes
3  12215000  7500         4          2        2      yes        no      yes
4  11410000  7420         4          1        2      yes       yes      yes

  hotwaterheating airconditioning  parking prefarea furnishingstatus
0              no             yes        2      yes        furnished
1              no             yes        3       no        furnished
2              no              no        2      yes   semi-furnished
3              no             yes        3      yes        furnished
4              no             yes        2       no        furnished

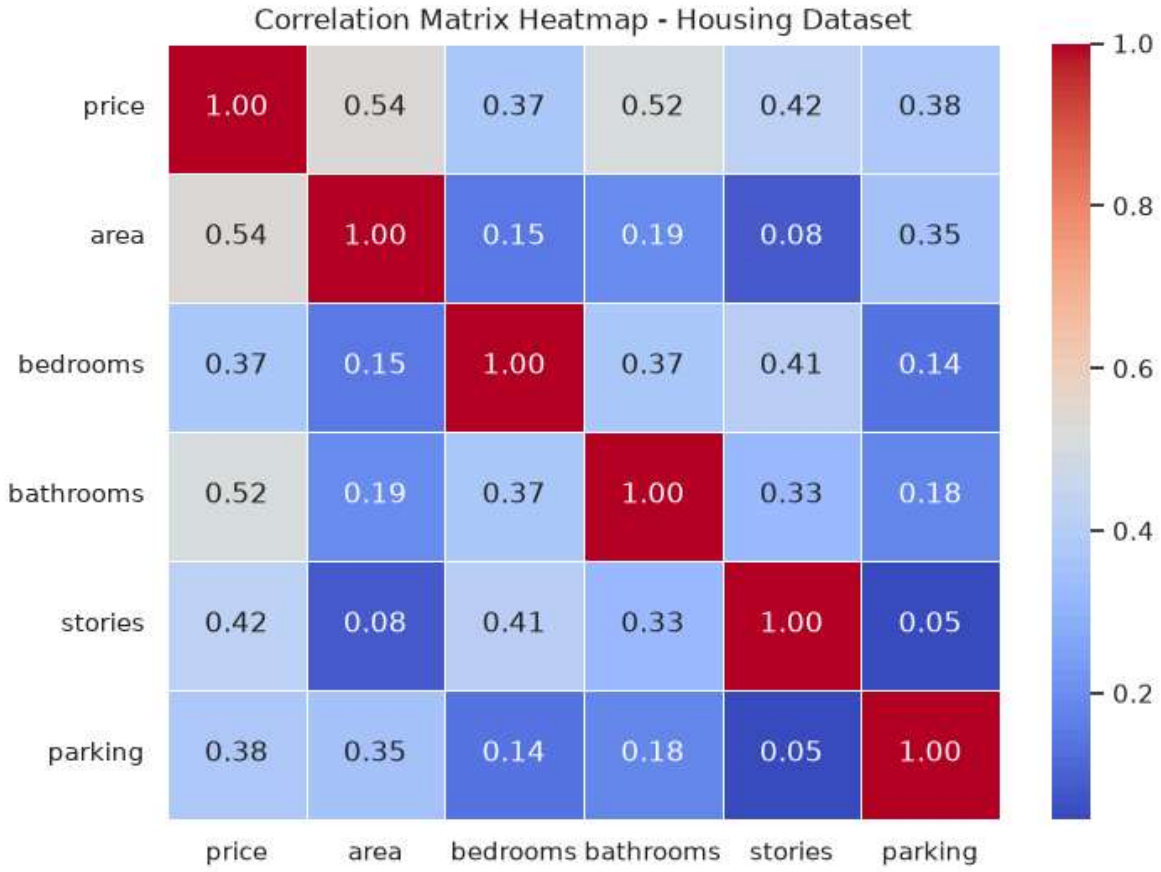
--- 2. DESCRIPTIVE STATISTICS (Numerical Features) ---
              price          area    bedrooms   bathrooms     stories  \
count  5.450000e+02    545.000000  545.000000  545.000000  545.000000
mean   4.766729e+06   5150.541284    2.965138    1.286239    1.805505
std    1.870440e+06   2170.141023    0.738064    0.502470    0.867492
min    1.750000e+06   1650.000000    1.000000    1.000000    1.000000
25%    3.430000e+06   3600.000000    2.000000    1.000000    1.000000
50%    4.340000e+06   4600.000000    3.000000    1.000000    2.000000
75%    5.740000e+06   6360.000000    3.000000    2.000000    2.000000
max    1.330000e+07  16200.000000    6.000000    4.000000    4.000000

          parking
count  545.000000
mean     0.693578
std      0.861586
min      0.000000
25%      0.000000
50%      0.000000
75%      1.000000
max      3.000000

Statistics for price:
  - Mean: 4766729.25
  - Median: 4340000.00
  - Mode: 3500000
  - Variance: 3498544355820.57
  - Standard Deviation: 1870439.62

Statistics for area:
  - Mean: 5150.54
  - Median: 4600.00
  - Mode: 6000
  - Variance: 4709512.06
  - Standard Deviation: 2170.14

--- 3. CORRELATION ANALYSIS ---
              price
price      1.000000
area       0.535997
bedrooms   0.366494
bathrooms  0.517545
stories    0.420712
parking    0.384394
```


Correlation Matrix Heatmap - Housing Dataset
price
area
bedrooms
bathrooms
stories
parking
1.00 0.54 0.37 0.52 0.42 0.38
0.54 1.00 0.15 0.19 0.08 0.35
0.37 0.15 1.00 0.37 0.41 0.14
0.52 0.19 0.37 1.00 0.33 0.18
0.42 0.08 0.41 0.33 1.00 0.05
0.38 0.35 0.14 0.18 0.05 1.00
price
area
bedrooms
bathrooms
stories
parking
1.0
0.8
0.6
0.4
0.2

```
--- 4. INDEPENDENT SAMPLE T-TEST ---
T-statistic: 4.4413
P-value: 1.1657e-05
Conclusion: Reject H0. There is a statistically significant difference in house p
rices based on basement availability.

--- 5. ONE-WAY ANOVA ---
F-statistic: 41.7823
P-value: 2.6832e-24
Conclusion: Reject H0. There is a statistically significant difference in house p
rices across different story counts.

--- 6. SIMPLE LINEAR REGRESSION MODEL ---
                            OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.287
Model:                            OLS   Adj. R-squared:                  0.286
Method:                 Least Squares   F-statistic:                     218.9
Date:                Mon, 24 Aug 2026   Prob (F-statistic):           7.39e-42
Time:                        08:11:37   Log-Likelihood:                -8551.2
No. Observations:                 545   AIC:                         1.711e+04
Df Residuals:                     543   BIC:                         1.712e+04
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept   2.387e+06   1.74e+05     13.681      0.000    2.04e+06    2.73e+06
area         461.9749     31.226     14.795      0.000     400.637     523.313
==============================================================================
Omnibus:                       92.668   Durbin-Watson:                   0.565
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              183.673
Skew:                           0.954   Prob(JB):                     1.31e-40
Kurtosis:                       5.108   Cond. No.                     1.44e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
[2] The condition number is large, 1.44e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```



--- 7. SUMMARY OF FINDINGS ---
- R-squared (R^2): 0.2873 (28.73% of price variance explained by area)
- Coefficient for Area: 461.97 (For every 1 sq.ft increase, price changes by this amount)
- Model p-value (F-statistic p-value): 7.3882e-42